DWS IQ × TECoSA — A Collaboration Overview

Prepared for Prof. Martin Törngren, KTH / TECoSA · from Risto Anton Päärni, Founder, CEO · Lifetime Oy · 14 July 2026

This is a short overview to let you judge — before we spend 30 minutes on a call — whether what we run in production connects to TECoSA's phase-2 trustworthiness themes. It is not a sales deck and not a doctoral proposal. It describes a deployed system, states plainly where the overlap with your work looks real, names the one concrete area where our platform could serve you as an empirical testbed, and draws an honest line around what we do *not* yet claim. Safety argumentation is your field, not mine; where this memo uses that vocabulary it is to locate our production artefact against your frameworks, not to claim authorship of the theory. If, having read it, you conclude the fit is not there, that is a useful answer and no time is wasted.

0. Background and Motivation

Two forces make the trustworthiness of agentic edge systems a live problem rather than a future one. First, regulation is arriving on a fixed timetable: the EU AI Act's Article 12 record-keeping obligation and Article 50 transparency duties, layered on top of NIS2 and the climate-disclosure regime (CBAM, CSRD), turn "can you show what your autonomous system did, and why" from good practice into a legal requirement for exactly the regulated industrial operators we serve. Second, the category itself is being defined in public right now — on-premises and edge "AI operating systems" for regulated industry are being announced by large vendors, most of them US-domiciled and therefore carrying CLOUD Act exposure at the vendor relationship. The EU-sovereign, verifiable variant of that category is not yet settled.

Our interest in reaching TECoSA is that the hard part of this is not building agents that act at the edge — they do, in our production system today — but making their actions *verifiable, legally defensible, and economically viable under real-time constraints*, and doing so as an engineering discipline rather than a compliance bolt-on. That is a methods-and-theory problem, which is your centre's ground, sitting directly on top of an applied system, which is ours. This memo is an attempt to test, honestly and quickly, whether those two grounds meet.

1. Production System Overview

Lifetime Oy (Finland) builds **DWS IQ**, a sovereign, on-premises / edge AI platform for regulated industrial operators. It runs autonomous and multi-agent workloads at the edge — the layer between industrial devices and the cloud, where inference must happen locally, under a fixed hardware budget, close to the process it serves. What runs in production today, stated factually rather than as a pitch:

- **18 governed agents** in the fleet, each with a defined role, its own trust score, and its own execution isolation.
- Coordination through an **MCP Hub** that implements **four production workflow patterns**, each exposing a defined endpoint with routing, error propagation, and result aggregation:
  - **Fan-Out** — one task dispatched to several agents in parallel, results aggregated.
  - **Sequential** — a pipeline where each agent's output is the next agent's input.
  - **Reflection** — an agent's output critiqued and revised by a second agent before it is accepted.
  - **Pipeline** — a staged flow with per-stage checkpoints and evidence capture.
- An **edge deployment (Aegis)** on NVIDIA DGX-class / GB10 Grace Blackwell hardware, running EU-resident inference on-premises inside the operator's environment.
- A small base of **paying industrial customers and pilot targets**, notably in steel (Outokumpu) and the construction sector — regulated settings where agent actions carry real operational consequence.

The distinguishing feature is not the models the platform runs but the **governance and evidence layer wrapped around every agent decision**. We describe ourselves as an "Agent Outcome Company": the deliverable is not a chatbot but an auditable outcome with an evidence trail. That layer — not the agent count — is what is relevant to TECoSA, and it is the subject of the rest of this memo.

2. Evidence Pipeline Architecture

The part of DWS IQ that maps to your safety-argumentation work is the trust-and-evidence layer. It turns a running agent — or a running fleet of agents — into an object that produces its own assurance evidence continuously. It has three components.

2.1 KYA (Know-Your-Agent) — trust scoring

Every agent carries a trust score derived from its provenance, its behaviour, and the constraints it operates under. In assurance-case terms, KYA is the quantified head of a safety argument — the claim "this agent is trustworthy enough to act here" — and it is only as strong as the evidence beneath it. We decompose the score conceptually into three sub-claims, each backed by a different part of the system:

- **Provenance** — is this agent what it says it is? (backed by attested identity)
- **Behavioural conformance** — has it stayed inside its declared envelope? (backed by the decision log)
- **Containment** — can it be stopped if it has not? (backed by the isolation guarantee and its measured teardown time)

2.2 Article-12 logging pipeline — the evidence chain

Every decision an agentic system takes at the edge is captured as structured, verifiable evidence. In effect, a logged decision record carries: a timestamp and monotonic sequence; the acting agent's attested identity; a reference to the inputs and the policy in force; the action taken and its outcome; the KYA score at the moment of action; and, for multi-agent flows, the causal link to the decision that licensed it. This is engineered to satisfy the EU AI Act's Article 12 record-keeping obligation, but the design intent is broader — an evidence chain an auditor, or a court, could follow end to end. **Regulatory alignment falls out of the evidence chain; it does not drive it.** That inversion — evidence first, compliance as a by-product — is the design stance we would most value your read on.

2.3 Firecracker MicroVM isolation — hardware-attested containment

Agents execute inside Firecracker MicroVMs with a measured sub-125 ms teardown (against the published AWS benchmark), so a misbehaving or compromised agent is contained at the hardware boundary rather than trusted to behave. This is what lets us treat **behaviour, not stated intent, as the admissible evidence** — for agentic systems the reasoning–behaviour gap makes intent unreliable, and hardware attestation is where a trustworthiness claim becomes falsifiable.

> **The evidence model, in your vocabulary.** The KYA score is the top-level claim; the three sub-claims above are *safety-argument fragments*, each with its own evidence and its own cost to produce. The score is what you get when they compose. What we do **not** have is a formal account of how those fragments compose into a sound top-level claim, nor of what each costs to produce under load — which is exactly the seam where your methodology would enter.

Multi-agent, not single-agent. Most industrial workloads are multi-agent: several agents acting, delegating, and constraining one another. Trustworthiness is then a property of the interaction, and the evidence chain must preserve inter-agent causality — which decision by agent A licensed which action by agent B. Our logging is built to keep that causal structure, which is where the reasoning–behaviour gap is sharpest and hardware-attested behavioural evidence is most clearly the only admissible ground.

3. Safety–Performance Trade-Off as a Live Engineering Problem

This is the specific area where our platform could be directly useful to your research rather than merely adjacent to it — and I want to frame it as a shared engineering problem, not a thesis topic.

The problem. Every increment of safety assurance has a price at the edge. Stronger evidence — deeper KYA checks, fuller Article-12 logging, tighter isolation boundaries — costs latency and compute, which are scarce in exactly the deployments where safety matters most. On the Aegis DGX deployment we face this tension daily, and we manage it today by judgement and measurement. **We do not have a formal model of it.**

Concretely, where the cost lands. Fuller logging adds I/O and serialization on the critical path; deeper KYA checks add inference and policy-evaluation time before an action is permitted; tighter isolation adds scheduling and teardown overhead. Individually small, together they move the end-to-end decision latency — and under a fixed real-time budget, past a point, the system must either slow down or start shedding assurance. Which of those it should do, and when, is a decision we currently make by operator judgement, not by a model that says where the efficient frontier lies.

Why your methodology is the closest fit. Your SAFECOMP work on safety-argument fragments and safety-performance trade-off analysis addresses exactly this, theoretically. An assurance case assembled from fragments, each with an explicit cost traded against performance under real-time constraints, is the formal object we are missing. What AD-EYE does for automated-driving *evaluation*, we would want to do for sovereign edge-AI *verification* — the same "evaluate before you trust" philosophy, one layer up, applied to agent fleets rather than vehicles.

4. Aegis DGX Deployment as Testbed

Aegis is our DGX-based edge configuration for dual-use / defence-adjacent industrial settings. It runs the same KYA + Article-12 evidence stack in a hardened, on-premises configuration where the constraints are real, not simulated: a fixed hardware budget, bounded latency, and agents that actuate on real industrial processes — so shedding an assurance check to save latency is a consequential decision, not a slider in a lab. That is the regime in which a safety-performance model earns its keep, and it is hard to reconstruct synthetically, which is much of why we think an instrumented production system is worth more to a verification researcher than another benchmark.

Element	Detail (to be confirmed against the live system before citation)
Edge hardware	NVIDIA GB10 Grace Blackwell (standard tier); hardened DGX-based configuration for Aegis
Inference	Models served locally, EU-resident; no US-hyperscaler dependency in the data path
Agent runtime	18 agents under KYA governance; four MCP-Hub workflow patterns (Fan-Out, Sequential, Reflection, Pipeline)
Tenant isolation	Firecracker MicroVM per agent; hardware-attested boundary
Kill-authority latency	Sub-125 ms MicroVM teardown (AWS benchmark)
Evidence	Structured per-decision logging engineered to EU AI Act Art. 12
Sovereignty	EU-resident; no CLOUD Act exposure in the deployment

4.1 What the testbed exposes for validation

For a trade-off model to be validated it needs instrumented dependent variables it can predict and be checked against. Aegis can expose, per decision and per workload:

- **End-to-end decision latency**, decomposed into inference / KYA-check / logging / isolation overhead — so the cost of each assurance fragment is separable, not lumped.
- **Compute and memory footprint of the assurance layer specifically**, as distinct from the useful work.
- **The KYA score and its sub-claims as a time series**, so a model can be tested against a moving assurance level rather than a snapshot.
- **Evidence-completeness under load** — where, as latency budgets tighten, checks or logging begin to be shed (the failure mode a safety-performance model should predict before it happens).
- **Isolation-event traces** — MicroVM teardown events, triggers, and timing, as the empirical ground for the containment fragment.

4.2 A proposed experimental protocol (for you to reshape)

Fix an industrial workload on Aegis. Sweep the assurance level — from minimal (identity check only) to maximal (full KYA + complete Article-12 logging + tightest isolation) — holding the workload constant, and at each point measure both the safety-evidence strength and the performance cost from §4.1. The output is an empirical safety-performance frontier for a real agentic edge system. Your methodology predicts where that frontier *should* sit and which fragment compositions are Pareto-efficient; the testbed says whether the prediction holds. If it does, you have a validated model on a production system; if it does not, the gap is itself a finding. Either way it is paper-shaped.

5. Collaboration Model

The shape we would want is the one you already run: **industry provides the deployment ground, the university provides the verification framework** — the pattern of your industrial PhD work with Scania, Alstom and Ericsson. We would come as a peer contributing an instrumented system, not as a supplier pitching a product.

	What each side brings	What each side gets
TECoSA / you	The trade-off methodology and verification framework; the exact SAFECOMP references to instantiate; the metric set	A live, instrumented production system to validate the model against; a paper-shaped result on real industrial data
Lifetime / us	The instrumented Aegis environment; engineering time to expose the measurement surface; the KYA specification	A formal backing for a trade-off we now manage by judgement; external verification of our assurance claims

5.1 A demonstrator sketch (three phases, for you to cut down)

- **Phase A — instrument & baseline.** We expose the §4.1 measurement surface for one agreed workload; you fix the assurance-level sweep and metrics. Deliverable: a clean baseline of safety-evidence strength vs. performance cost, no modelling claims yet.
- **Phase B — instantiate the model.** You express your fragment / trade-off methodology against the baseline: which fragments, what each costs, where the Pareto-efficient compositions sit. Deliverable: a model with explicit, testable predictions.
- **Phase C — validate & write.** We run the predicted operating points on Aegis under load; you check prediction against measurement. Deliverable: a trade-off model validated (or falsified) on a production sovereign edge-AI system, with failure modes named. Authorship and framing are yours to lead; our contribution is the system and the data.

5.2 Practical terms we would propose up front

So that the engagement is clean rather than improvised, we would want to agree early on:

- **Data.** Measurement data (latencies, footprints, scores) is shareable; raw customer process data stays inside the operator's environment and is not exported. Workload owners' consent governs anything customer-specific.
- **Publication.** Academic publication rights sit with your side; we ask only for a factual read of any system description and a short window to flag genuinely sensitive operational detail before submission — not editorial control.
- **IP.** Your methodology stays yours; our platform and KYA specification stay ours; jointly produced results (the validated model, the datasets) are jointly owned or licensed as we agree.
- **Scope discipline.** One workload, one trade-off, one paper first. Breadth later, only if the first result holds.

5.3 What we do NOT claim (the honest boundary)

A collaboration works only if the boundary is honest. The platform does **not** yet cover: a formal soundness proof for the KYA score; a quantified cost-of-safety curve; tested legal admissibility of the evidence chain; portability across other runtimes and model families; external calibration of the trust score against a ground truth; or bounds on multi-agent emergent behaviour. None of these is hidden — each is an open problem, and several are exactly what a verification collaboration would attack. Equally honestly, the overlap could fail: the frontier may be dominated by mundane inference cost rather than fragment structure; agentic behaviour may be too unlike connected automated vehicles for the fragment composition to transfer; or a result validated only on Aegis may not generalise. We would rather surface these up front than oversell a fit that does not survive contact with your standards.

6. Specific Overlap with TECoSA Phase-2 Themes & SAFECOMP

From the outside, the overlap with TECoSA's phase-2 trustworthiness work looks real on three counts:

- **Trustworthy edge computing as the headline.** TECoSA's load-bearing nouns — trustworthy, edge, systems — are the three our platform is built around. We are the applied/commercial instantiation of the layer your centre studies as methods and theory. Your position paper on trustworthy edge computing in industrial applications is, from where I sit, the closest framing in print to what we run in production.
- **Safety-argument fragments and the trade-off.** The KYA score decomposes into provenance / conformance / containment fragments (§2); your SAFECOMP work on fragments and safety-performance trade-off analysis is the methodology that would give those fragments a formal footing and a cost model. That is the specific overlap of §3–§4, and the reason I am writing to you rather than to the field in general.
- **The AD-EYE parallel.** Where AD-EYE is the controlled environment on which driving behaviours are evaluated before they are trusted, Aegis would be the instrumented environment on which an assurance *policy* — how much evidence, produced how, at what cost, buys how much trustworthiness — is validated before it is relied on. Same philosophy; the object under test moves from a vehicle's behaviour to an agent fleet's assurance.

> **On references, honestly.** I have named your position paper and SAFECOMP work by topic rather than pretending to a line-by-line command of the proofs — safety argumentation is your field, and I would rather get the exact references and their claims right *with you* on the call than mis-state them here. What I can speak to without qualification is the production system and the tension it lives; the theory is where I would be learning from you.

If, reading this, you conclude the fit is there, the natural next step is 30 minutes to agree which SAFECOMP references and trade-off model to instantiate and the scope of a first demonstrator. If you conclude it is not, that is an equally useful answer.

7. The Next Step

Concretely, I am asking for 30 minutes. In that call I would hope to establish three things: your read on whether §3–§4 (the safety–performance trade-off overlap) connects to TECoSA's phase-2 work, or not; if it does, the exact SAFECOMP references and trade-off model you would want to instantiate, so I can prepare the Aegis testbed to match; and the scope of a first demonstrator small enough to produce one clean result. I bring the system and its measurement surface; you bring the judgement on whether the fit is real. If the answer is no, it is a fast and useful no, and I will have taken little of your time.

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Appendix A — Glossary

DWS IQ

Lifetime Oy's sovereign, on-premises / edge AI platform for regulated industrial operators.

Aegis

The DGX-based, hardened edge deployment of DWS IQ for dual-use / defence-adjacent industrial settings; the proposed testbed.

KYA (Know-Your-Agent)

The trust-scoring layer; a quantified assurance claim per agent, decomposed into provenance, behavioural conformance, and containment.

Article 12

The EU AI Act record-keeping / automatic-logging obligation for high-risk AI systems; the design target of the evidence pipeline.

Article 50

The EU AI Act transparency obligation (context, not the subject of this memo).

Firecracker MicroVM

Lightweight virtualization providing per-agent hardware-attested isolation with sub-125 ms teardown (AWS benchmark).

MCP Hub

The coordination layer implementing the four production workflow patterns (Fan-Out, Sequential, Reflection, Pipeline).

Assurance case / safety-argument fragment

A structured argument that a system is acceptably safe, assembled from fragments each carrying its own evidence — the framing from your SAFECOMP work that this memo maps KYA onto.

NIS2

EU cybersecurity directive for essential and important entities; part of the regulatory frame our industrial customers operate under.

References & Sources

Provided for grounding; specific academic references to be confirmed with Prof. Törngren before the call rather than cited from memory here.

1. M. Törngren et al., position paper on trustworthy edge computing in industrial applications (TECoSA / KTH) — exact citation to confirm.
2. M. Törngren et al., SAFECOMP work on safety-argument fragments and safety-performance trade-off analysis for connected automated vehicles — exact citation(s) and year(s) to confirm (DBLP / SAFECOMP proceedings).

3. AD-EYE automated-driving evaluation testbed (KTH) — referenced as the methodological analogue.
4. TECoSA — Trustworthy Edge Computing Systems and Applications, Vinnova competence centre hosted by KTH (phase 2).
5. EU AI Act, Regulation (EU) 2024/1689 — Articles 12 (record-keeping) and 50 (transparency).
6. AWS Firecracker MicroVM — teardown / isolation benchmark (source of the sub-125 ms figure).

Appendix B — Indicative per-decision measurement fields

What a Phase-A baseline would expose, per logged decision, so a trade-off model has an instrumented right-hand side. Field names are illustrative; the exact schema is agreed in Phase A.

Field	What it captures	Why the model needs it
decision_id / seq	Unique id + monotonic sequence	Orders events; anchors the causal graph
agent_id (attested)	Which agent acted, with attested identity	Provenance fragment
inference_ms	Model inference time for the decision	Baseline performance cost (not assurance)
kya_check_ms	Time spent evaluating the KYA checks	Cost of the trust-scoring fragment
logging_ms	Time to serialize and persist the evidence record	Cost of the evidence fragment
isolation_ms	Scheduling / teardown overhead attributable to isolation	Cost of the containment fragment
kya_score + sub-claims	Trust score and its provenance / conformance / containment parts	The assurance level being traded
evidence_complete	Whether full evidence was captured, or shed under load	Detects the shedding failure mode
causal_parent	The decision that licensed this one (multi-agent flows)	Preserves inter-agent causality

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Confidential — collaboration correspondence, Lifetime Oy. System facts (hardware SKUs, agent and workflow-pattern counts, MicroVM benchmark, customer status) are stated to the best of current knowledge and should be confirmed against the live system before external citation. Exact SAFECOMP and position-paper references to be verified prior to the call.